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Random Variables II: Discrete Distributions

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Stat 141 | Fall 2026

Monday, Week 11

Goals for Today

- Brief review of expected value and variance
- Introduce the **Bernoulli distribution**
- Introduce the **Binomial distribution**

Review

Review: Expected Value, Variance, and Standard Deviation

The **expected value** (or mean) of a discrete random variable X is

$$\mu = E[X] = \sum_{i=1}^{n} x_i P(X = x_i)$$

where $x_1, \ldots, x_n$ are *all the values X could potentially take*.

The **variance** and **standard deviation** of a discrete random variable X with mean $E(X) = \mu$ is

$$\sigma^2 = \text{Var}(X) = \sum_{i=1}^{n} (x_i - \mu)^2 P(X = x_i)$$

$$\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}$$

Review: Rules for Expected Value and Variance

Theorem: Expectation of Sum

Let X and Y be random variables. Then

$$E(X+Y) = E(X) + E(Y)$$

Theorem: Scalar Multiplication with Expectation

Let X be a random variable, and let c be a number. Then

$$E(cX) = cE(X)$$

Theorem: Variance of Sum for Independent Random Variables

Let X and Y be random variables. Additionally, let X and Y be *independent*. Then

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

Theorem: Scalar Multiplication with Variance

Let X be a random variable, and let c be a number. Then

$$\text{Var}(cX) = c^2\text{Var}(X)$$

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Recall: One Coin Flip

Consider a random variable X which is the number of heads in a single coin flip.

Q: What are the possible values for X? What are the probabilities each value of X?

- $X = 1$ (H) with $P(X=1) = \frac{1}{2}$
- $X = 0$ (T) with $P(X=0) = \frac{1}{2}$

Q: Compute the expected value and variance for X (the number of heads in a single coin flip)

$$E[X] = (0)\frac{1}{2} + (1)\frac{1}{2} = \frac{1}{2}$$

$$Var[X] = (0-1/2)^2\frac{1}{2} + (1-1/2)^2\frac{1}{2} = \frac{1}{4}$$

This distribution has a name!

- X is called a **Bernoulli** random variable!

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Specific Named Random Variables

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Specific Named Random Variables

- There is a vast array of random variables out there.
- But there are a few particular ones that we will find useful.
  - Because these ones are used often, they have been given names.
- Will identify these named RVs using the following format:

$$X \sim \text{Name(values of key parameters)}$$

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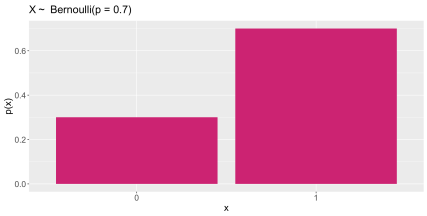
Bernoulli Random Variables

$X \sim$ Bernoulli (p)

$$X = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

Important parameter:

$$p = \text{probability of success} = P(X = 1)$$



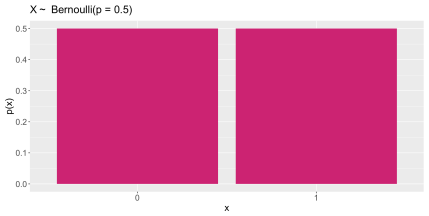
Distribution:

x	0	1
$p(x)$	1 - p	p

Bernoulli Random Variables

$X \sim$ Bernoulli $(p = 0.5)$

$$X = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$



Distribution:

x	0	1
$p(x)$	0.5	0.5

Bernoulli Random Variables

$X \sim$ Bernoulli (p)

$$X = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

Distribution:

x	0	1
$p(x)$	1 - p	p

Question: What's the expected value of a Bernoulli(p) random variable?

01:30

Bernoulli Random Variables

$X \sim$ Bernoulli (p)

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Distribution:

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Question: What's the expected value of a Bernoulli(p) random variable?

$$\mu = \sum xp(x)$$

Bernoulli Random Variables

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$$X = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

Distribution:

x	0	1
$p(x)$	1 - p	p

Question: What's the expected value of a Bernoulli(p) random variable?

$$\begin{aligned} \mu &= \sum xp(x) \\ &= 1 * p + 0 * (1 - p) \\ &= p \end{aligned}$$

Bernoulli Random Variables

$X \sim$ Bernoulli (p)

$$X = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

Distribution:

x	0	1
$p(x)$	1 - p	p

Question: What's the standard deviation of a Bernoulli(p) random variable?

02:00

Bernoulli Random Variables

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Distribution:

x	0	1
$p(x)$	1 - p	p

Question: What's the standard deviation of a Bernoulli(p) random variable?

$$\sigma = \sqrt{\sum (x - \mu)^2 p(x)}$$

Bernoulli Random Variables

$X \sim$ Bernoulli (p)

$$X = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

Distribution:

x	0	1
$p(x)$	1 - p	p

Question: What's the standard deviation of a Bernoulli(p) random variable?

$$\begin{aligned} \sigma &= \sqrt{\sum (x - \mu)^2 p(x)} \\ &= \sqrt{(1 - p)^2 * p + (0 - p)^2 * (1 - p)} \end{aligned}$$

Bernoulli Random Variables

$X \sim$ Bernoulli (p)

$$X = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

Distribution:

x	0	1
$p(x)$	1 - p	p

Question: What's the standard deviation of a Bernoulli(p) random variable?

$$\begin{aligned} \sigma &= \sqrt{\sum (x-\mu)^2 p(x)} \\ &= \sqrt{(1-p)^2 * p + (0-p)^2 * (1-p)} \\ &= \sqrt{p(1-p)} \end{aligned}$$

Activity: Many Bernoulli random variables

What about 23 coin tosses?

We can model one fair coin toss as a $\textbf{Bernoulli}(0.5)$. How should we think about tossing the coin many (ex. 23) times?

Let Z count the number of heads in $\mathbf{23}$ independent coin tosses.

Q: Let Z_i represent the ith coin toss. Can we write Z in terms of the Z_is?

Q: What is $E(Z)$? (*Hint: you do not need to write out the probability distribution of Z.*)

Q: What is $\text{Var}(Z)$?

Q: Suppose X represents the number of heads in n independent coin tosses. What is $E(X)$ and $\text{Var}(X)$?

05:00

Activity: Answers

Let Z be the number of heads in $\mathbf{23}$ independent *fair* coin tosses.

Q: Let Z_i represent the ith coin toss. Can we write Z in terms of the Z_is?

$$Z = Z_1 + Z_2 + \cdots + Z_{23}$$

Q: What is $E(Z)$?

$$E(Z) = E(Z_1) + \cdots + E(Z_{23}) = 0.5 + \cdots + 0.5 = 23(0.5)$$

Q: What is $\text{Var}(Z)$?

$$\text{Var}(Z) = \text{Var}(Z_1) + \cdots + \text{Var}(Z_{23}) = 0.25 + \cdots + 0.25 = 23(0.25)$$

Q: Suppose X represents the number of heads in n independent coin tosses. What is $E(X)$ and $\text{Var}(X)$?

- $E(X) = nE(X_1) = (n)(0.5)$
- $\text{Var}(X) = n\text{Var}(X_1) = (n)(0.25)$

n "weighted coin tosses": Expected Value and Variance

Let X be the number of successes in n independent Bernoulli(p) random variables.

Q: What are $E(X)$ and $\text{Var}(X)$?

We know, from the Bernoulli distribution, that if X_1 is the number of heads in a *single* weighted coin flip with success probability p,

$$E(X_1) = p \quad \text{and} \quad \text{Var}(X_1) = p(1-p)$$

n "weighted coin tosses": Expected Value and Variance

Let X be the number of heads in n independent Bernoulli(p) random variables.

Let X_1 = Heads in 1st toss, . . . , X_n = Heads in nth toss:

$$X = X_1 + \cdots + X_n = \sum_{i=1}^{n} X_i$$

$$E(X) = E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E(X_i) = \sum_{i=1}^{n} p = np$$

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \text{Var}(X_i) = \sum_{i=1}^{n} p(1-p) = np(1-p)$$

What about the distribution of n weighted coin tosses?

We'll start with the unweighted case, i.e., $p = 0.5$

n coin toss probabilities?

We've calculated Expectation and Variance for the number of heads in n coin tosses.

- But what about the probability that there are k heads in the n tosses?

For example, What's the probability of $k = 2$ heads in $n = 3$ coin tosses?

- Let's break down the problem: What are the possible outcomes?

$$\text{TTT} \implies X = 0 \qquad \text{TTH or THT or HTT} \implies X = 1$$
$$\text{HHH} \implies X = 3 \qquad \text{THH or HTH or HHT} \implies X = 2$$

- There are 8 possible sequences from 3 coin flips, each with equal probability.

The random variable X has 4 possible outcomes: 0, 1, 2, or 3. Those probabilities are:

$$P(X=0) = \frac{1}{8} \qquad P(X=1) = \frac{3}{8}$$
$$P(X=3) = \frac{1}{8} \qquad P(X=2) = \frac{3}{8}$$

3 coin tosses probabilities

The random variable X has 4 possible outcomes: 0, 1, 2, or 3. Those probabilities are:

$$P(X=0)=\frac{1}{8} \qquad P(X=1)=\frac{3}{8}$$
$$P(X=3)=\frac{1}{8} \qquad P(X=2)=\frac{3}{8}$$

Notice that the probability of k heads in n coin flips is

$$P[X=k] = (\text{Number of ways to get k heads}) \times (\text{Prob of each flip sequence})$$

For example, we found that in $n=3$ coin tosses,

- 3 ways to get 2 heads
- Each sequence of 3 coin flips has probability $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$
- Thus, $P(X=2) = \frac{3}{8}$

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Generalization 1

First: **What if n and k are bigger?**

- In this case, it's hard to individually count all the different flip sequences
- Instead, we can rely on a **handy formula**: The number of ways to get k "successes" in n "trials" is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

We read this as **"n choose k"**

The exclamation point is a **factorial**:

$$0! = 1$$
$$1! = 1$$
$$2! = 2 \times 1 = 2$$
$$3! = 3 \times 2 \times 1 = 6$$
$$4! = 4 \times 3 \times 2 \times 1 = 24$$

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Generalization 1

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$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

We read this as **"n choose k"**

So in our example of 2 heads in 3 flips:

$$\#\text{ of ways to choose 2 things out of 3 things} = \binom{3}{2} = \frac{3!}{2! \cdot 1!} = \frac{3 \cdot 2}{2} = 3$$

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Generalization 2

Second: **What if the probability of "success" changes?**

- In other words, how can we generalize to the sum of Bernoulli(p) case from the sum of the Bernoulli($p=0.5$) case?
- In this case, the probability of getting *a particular sequence* of k "successes" in n **independent** trials is:

$$p^k \times (1-p)^{n-k}$$

- There are k "successes", each with probability $p \rightarrow p^k$
- There are $n-k$ "failures", each with probability $1-p \rightarrow (1-p)^{n-k}$

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Binomial Distribution: Putting it all together

Thus, the probability of observing k successes in n independent Bernoulli trials with probability of success p is

$$P[X = k] = \binom{n}{k} p^k (1-p)^{n-k}$$

In this case, we say $X \sim \text{Binomial}(n,\ p)$. Or, in words: X follows a **binomial distribution** with n trials each with p probability of success

How to tell if your distribution is a Binomial?

- Number of trials, n, must be **fixed**.
- Each trial must be **Bernoulli** and independent, with success probability p (p must stay the same for each trial!).
- We must be counting the **number of successes**.

Properties

Expectation and Variance of Binomial

If $X \sim \text{Binomial}(n,\ p)$, then:

$$E(X) = np$$
$$Var(X) = np(1-p)$$

Note that we derived this earlier before we knew what a Binomial was, just by using Theorems about sums of random variables!

Calculating Binomial Probabilities: `dbinom()`

If X is a binomial random variable with n trials and probability of success p, we can use R to calculate the probabilities with the `dbinom()` function

- e.g., $P(X = 2)$ where $n = 3$ and $p = 0.5$ (2 heads in 3 fair coin tosses)

```
dbinom(x = 2, size = 3, prob = 0.5)
```

```
[1] 0.375
```

- e.g., $P(X = 10)$ where $n = 15$ and $p = 0.7$

```
dbinom(x = 10, size = 15, prob = 0.7)
```

```
[1] 0.2061304
```

- `dbinom()` takes in an observable value of a Binomial random variable, and returns the probability of observing it.

Calculating Binomial Probabilities: `pbinom()`

We can also calculate $P(X \leq k)$ with `pbinom()`

- e.g., $P(X \leq 2)$ where $n = 3$ and $p = 0.5$ (2 or fewer heads in 3 fair coin tosses)

```
pbinom(q = 2, size = 3, prob = 0.5)
```

```
[1] 0.875
```

Can also calculate $P(X > k)$ by doing `1 - pbinom()`:

- e.g., $P(X > 2)$ where $n = 3$ and $p = 0.5$ (more than 2 heads in 3 fair coin tosses)

```
1 - pbinom(q = 2, size = 3, prob = 0.5)
```

```
[1] 0.125
```

- e.g., $P(X \geq 2)$ where $n = 3$ and $p = 0.5$

```
1 - pbinom(q = 1, size = 3, prob = 0.5)
```

```
[1] 0.5
```

- Third bullet: why q = 1? `pbinom()` takes in an observable value of a Binomial random variable, and returns the probability of observing **that value OR something smaller**.

Recap

If $X \sim$ **Bernoulli** (p) (ex. one coin flip), then:

- $\mathbb{E}[X] = p$ and $Var(X) = p(1-p)$
- The distribution of X is:

x	0	1
$p(x)$	$1-p$	p

If $X \sim$ **Binomial**$(n,\ p)$ (i.e. X represents the number of successes in n independent Bernoulli trials with probability of success p), then:

- $\mathbb{E}[X] = np$ and $Var(X) = np(1-p)$
- The distribution of X is:

$$P[X = k] = \binom{n}{k} p^k (1-p)^{n-k}$$

- We can calculate Binomial probabilities using `dbinom()` and `pbinom()`.

Next time

- Another binomial random variable activity
- Continuous random variables
- The Central Limit Theorem!